

MAUVE–MUSE: When Metallicity Follows or Fights Star Formation

A Mass-Dependent Inversion in Virgo Spirals

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Billboard Result

At fixed local Σ_* , the resolved $Z_{\text{gas}}\text{--}\Sigma_{\text{SFR}}$ coupling **flips sign**:
dilution-driven anti-correlation at low Σ_* $\Rightarrow$ enrichment-driven positive correlation at high Σ_* ,
with a transition near $\log_{10}(\Sigma_*/M_\odot \text{ kpc}^{-2}) \approx 7.5\text{--}8.0$.

Why this matters + Data

- Resolved scaling relations constrain the local balance of **enrichment vs dilution**.
- MAUVE–MUSE Virgo spirals at ~ 100 pc resolution (star-forming spaxels).
- Fiducial ensemble: 13 galaxies (excluding outlier NGC 4383).
- Conservative star-forming mask (BPT-based using [N II] and [S II]).
- Fiducial metallicity: O3N2–M13 (poster focuses on the main results).

How we isolate the secondary coupling (poster version)

Remove the dominant dependence on Σ_* :

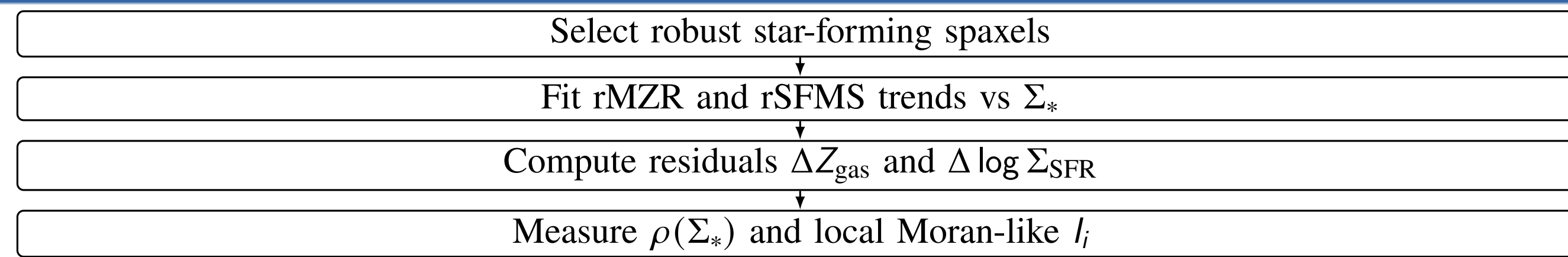
$$\Delta Z_{\text{gas}} \equiv Z_{\text{gas}} - \langle Z_{\text{gas}} \mid \Sigma_* \rangle, \quad \Delta \log \Sigma_{\text{SFR}} \equiv \log \Sigma_{\text{SFR}} - \langle \log \Sigma_{\text{SFR}} \mid \Sigma_* \rangle.$$

Then quantify the coupling as a function of Σ_* :

$$\rho(\Sigma_*) \equiv \text{corr}(\Delta Z_{\text{gas}}, \Delta \log \Sigma_{\text{SFR}} \mid \Sigma_*).$$

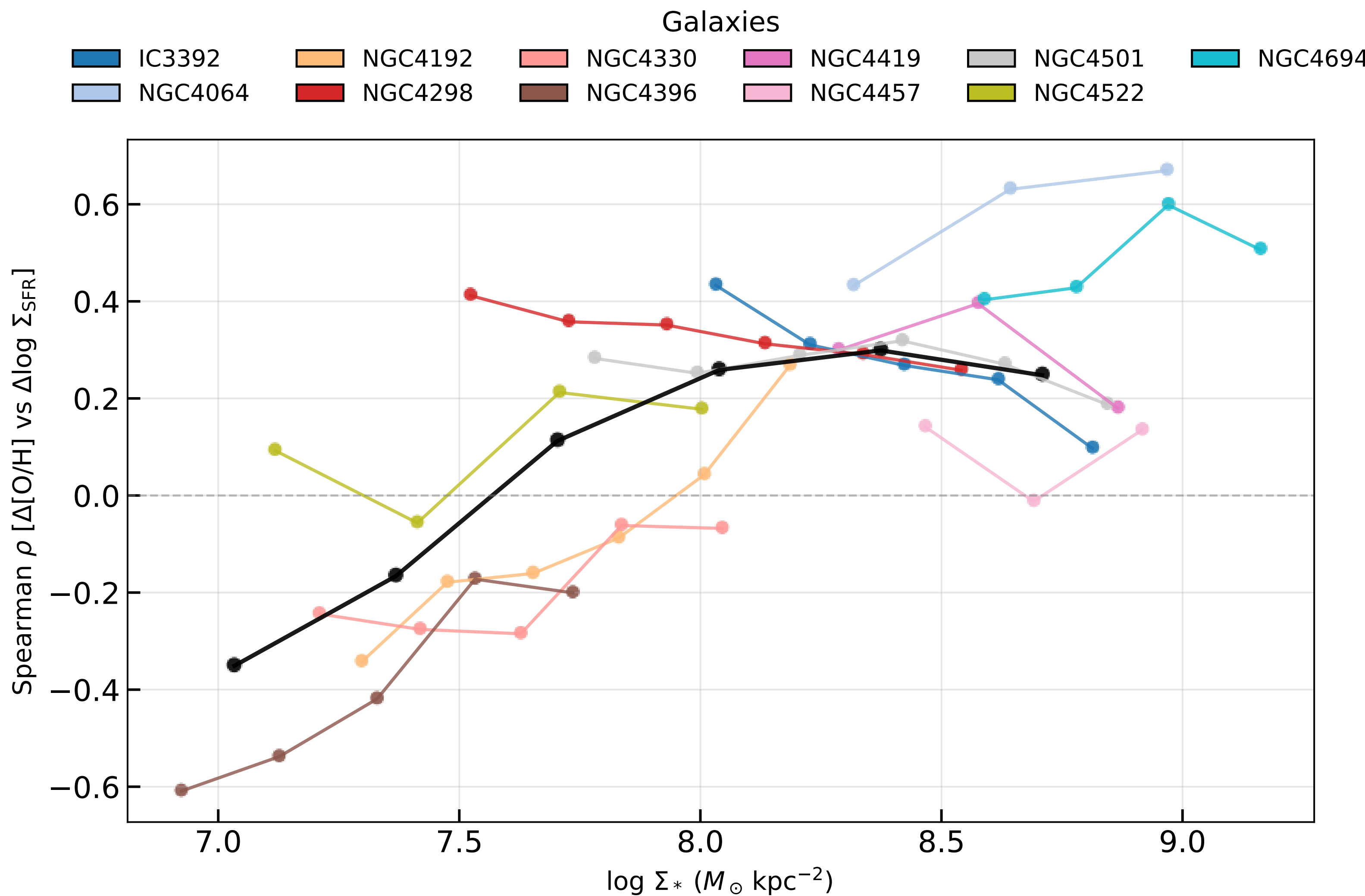
Finally test whether it is spatially coherent using a local Moran-like index I_i (“patches” of correlation / anti-correlation, not just a global statistic).

Analysis workflow + key numbers



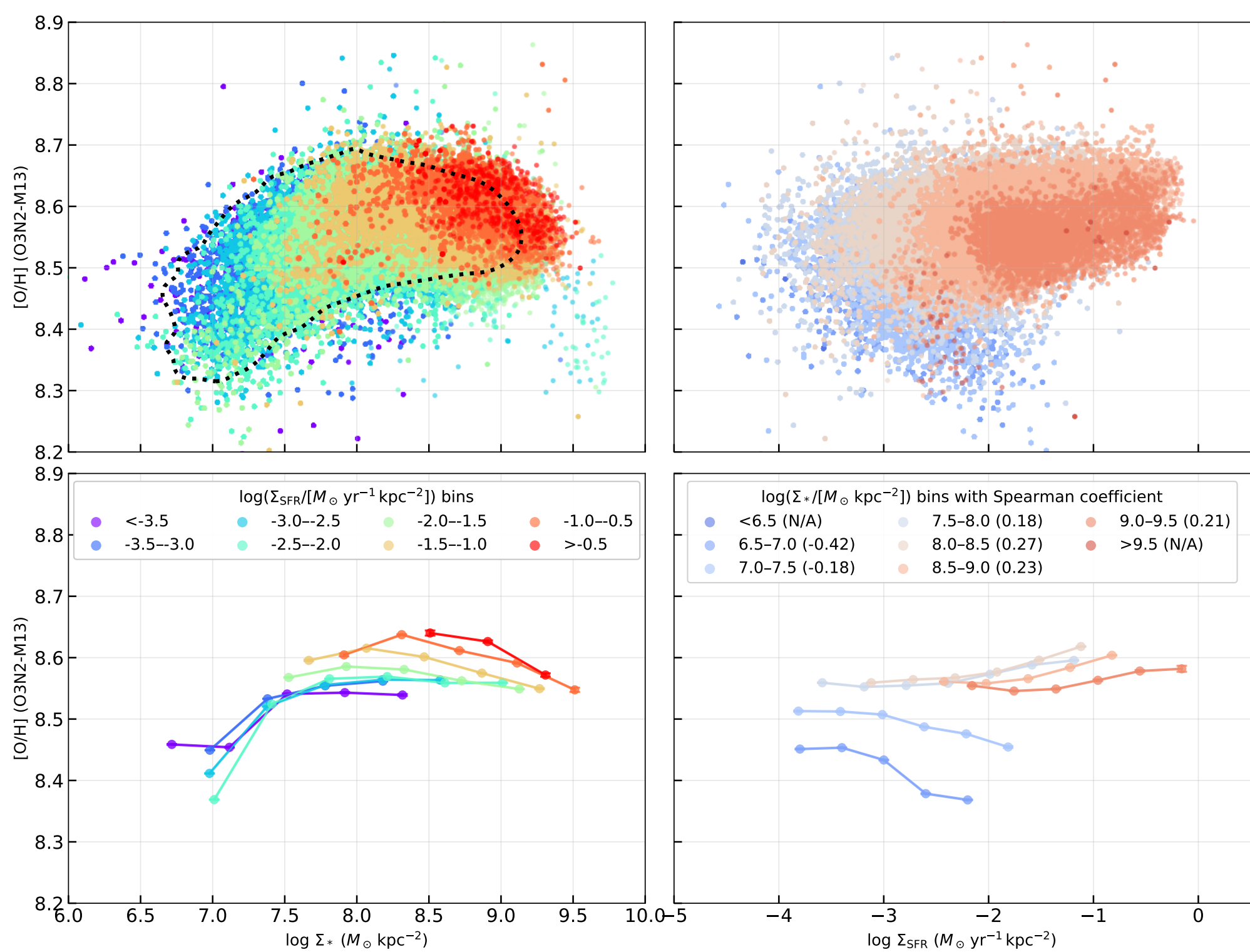
- Lowest- Σ_* bin reaches $\rho \approx -0.41$.
- High- Σ_* bins become positive (up to $\rho \sim +0.3$).
- NGC 4192 shows an internal radial mix of correlated / anti-correlated patches.

Main result: mass-dependent sign flip in residual $\Delta Z_{\text{gas}}\text{--}\Delta \log \Sigma_{\text{SFR}}$ coupling



Read in 5 seconds: $\rho(\Sigma_*)$ transitions smoothly from **negative (dilution-dominated)** to **positive (enrichment-dominated)**, with a zero-crossing near $\log_{10}(\Sigma_*) \sim 7.5\text{--}8.0$.

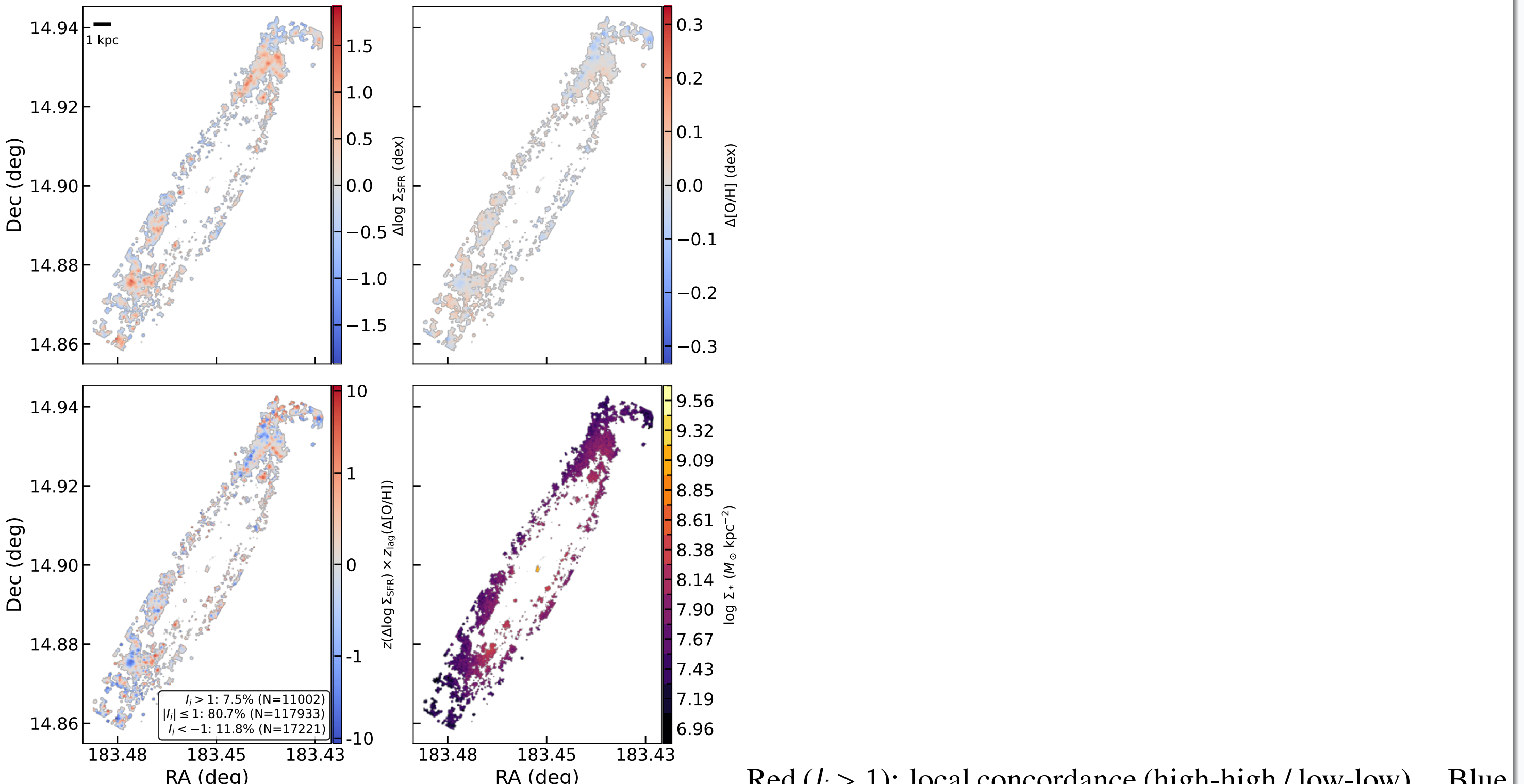
Baseline context: the rMZR (coloured by Σ_{SFR})



The rMZR is well-defined; the key is

the **secondary dependence** on Σ_{SFR} , which changes sign across Σ_* bins.

Spatial proof: local Moran-like map in NGC 4192



Red ($I_i > 1$): local concordance (high-high / low-low). Blue ($I_i < -1$): local discordance (high-low). Coexistence implies the inversion is **local and physical**.

Interpretation: local gas-regulator explains the sign flip

$$\frac{d\Sigma_g}{dt} = \Sigma_\Phi - (1 - R)\Sigma_{\text{SFR}} - \Sigma_{\text{out}}, \quad \frac{dZ}{dt} = \frac{1}{\Sigma_g} \left[y(1 - R)\Sigma_{\text{SFR}} + (Z_\Phi - Z)\Sigma_\Phi \right].$$

